

1. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

```
snigdha@Snigdha:~$ ./practical2
Usage: ./practical2 <source_file> <destination_file>
snigdha@Snigdha:~$ cat practical2.c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

#define BUFFER_SIZE 1024

int main(int argc, char *argv[])
{
    int source_fd, dest_fd;
    ssize_t bytesRead, bytesWritten;
    char buffer[BUFFER_SIZE];

    // Check command-line arguments
    if (argc != 3)
    {
        printf("Usage: %s <source_file> <destination_file>\n", argv[0]);
        return 1;
    }

    // Open source file (read only)
    source_fd = open(argv[1], O_RDONLY);

    if (source_fd < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    // Create/Open destination file
    dest_fd = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (dest_fd < 0)
    {
        perror("Error opening destination file");
        close(source_fd);
        return 1;
    }

    // Read from source and write to destination
    while ((bytesRead = read(source_fd, buffer, BUFFER_SIZE)) > 0)
    {
        bytesWritten = write(dest_fd, buffer, bytesRead);

        if (bytesWritten != bytesRead)
        {
            perror("Write Error");
            close(source_fd);
            close(dest_fd);
            return 1;
        }
    }

    if (bytesRead < 0)
    {
        perror("Read Error");
    }

    close(source_fd);
    close(dest_fd);

    printf("File copied successfully.\n");

    return 0;
}
```

```
snigdha@Snigdha:~$ nano practical2.c
snigdha@Snigdha:~$ gcc practical2.c -o practical2
snigdha@Snigdha:~$ ./practical2
Usage: ./practical2 <source_file> <destination_file>
snigdha@Snigdha:~$ |
```

```
snigdha@Snigdha:~$ gcc practical2.c -o practical2
snigdha@Snigdha:~$ nano source.txt
snigdha@Snigdha:~$ ./practical2 source.txt destination.txt
File copied successfully.
snigdha@Snigdha:~$ cat destination.txt
Hello Linux Operating System
This is a sample file.
OS Practical 2
snigdha@Snigdha:~$ ls -l source.txt destination.txt
-rw-r--r-- 1 snigdha snigdha 67 Aug  5 04:04 destination.txt
-rw-r--r-- 1 snigdha snigdha 67 Aug  5 04:03 source.txt
snigdha@Snigdha:~$ |
```